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SEAT No. :

PE-4293

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S.E.-Artificial Intelligence & Machine Learning

DATABASE MANAGEMENT SYSTEM

(2019 Pattern) (Semester - IV) (218554)

Time : 2½ Hours] *[Max. Marks : 70*

Instructions to the candidates :

1) ***Neat diagrams must be drawn wherever necessary.***
2) ***Figures to the right side indicate full marks.***
3) ***Assume Suitable data if necessary.***

Q1) a) Describe the circumstances in which you would choose to use embedded SQL rather than using SQL alone or using only a general purpose programming language. Compare dynamic and embedded SQL with suitable example. **[8]**

b) Consider following database: **[6]**

Student(rollno,name,address)

Subject(subjectcode,subjectname)

Marks(rollno,subjectcode,marks)

Write following queries in SQL

1) Find average marks of each student along with name of student.
2) Find how many student have failed in subject DBMS.

c) Differentiate between : WHERE and HAVING clauses in SQL. **[4]**

OR

Q2) a) What is the need of embedded SQL? Explain dynamic SQL with example. **[6]**

b) Explain various set operation in SQL with example.. **[6]**

c) What is view? Given the following schema **[6]**

Employee(eno,address,dno,salary)

Department(dno,dname,location,manager)

Using SQL generate a view to display information department wise as dno, total number of employee, total salary, average salary.

Q3) a) Define BCNF? How does it differ from 3NF?why is it consider a stronger form of 3NF? **[7]**

b) Given a relation schema R = (A,B,C,D,E) and function dependency as $A \to C$, $C \to D$, $CE \to A$, $B \to C$, $DE \to C$. Relation R is decomposed into r1 = AD, r2 = AB, r3 = BE, r4 = CDE, r5 = AE. Decide this decomposition is lossy or lossless ? Justify **[6]**

c) Write a note on Materialization. **[4]**

OR

Q4) a) Compute the closure of following set F of functional dependencies for relation schema R(A,B,C,D,E) **[6]**

$A \to BC$, $CD \to E$, $B \to D$, $E \to A$

Also find candidate key.

b) Show that with suitable example: if a relation schema is in BCNF, then it is also in 3NF **[6]**

c) Explain update anomalies with example. **[5]**

Q5) a) Give test for conflict serializability. Check whether following schedule is conflict serializable. **[6]**

T1	T2
Read(A)	
Write(A)	
	Read(A)
	Write(A)
Read(B)	
Write(B)	
	Read(B)
	Write(B)

b) What is meant by transaction? Explain transaction model with state diagram. **[6]**

c) Explain time-stamp based concurrency control. **[6]**

OR

Q6) a) Since every conflict serializable schedule is view serializable, why do we emphasize conflict serializability rather than view serializability. **[6]**

b) When do deadlock happen, how to recover if deadlock takes place? **[6]**

c) Explain deferred database modification and immediate database modification and their differences in the context of recovery. **[6]**

Q7) a) State which database system architecture you will prefer for following application **[6]**

i) Banking system

ii) Airline management

iii) College admission system

b) Explain speed- up and scale-up .issue with respect to parallelism. **[6]**

c) What are the requirement of mobile databases? List existing mobile databases. **[5]**

OR

Q8) a) What is fragment of a relation? what are the main type of fragmentation? Why is fragmentation a useful context in distributed database design. **[6]**

b) Draw and explain architecture of parallel Databases **[6]**

c) Write a note on XML databases. **[5]**

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